

Logistic regression

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1 Introduction

There are many method to classify elements thanks to features given.

We are looking at the regression method, and study the performances of this method

2 Methodology

- X : a set of features ($n \times d$)
- y : a set of labels for the given features ($n \times h$)
- C : the set of all the classes

2.1 Binary case

In this case, $h = 1$.

We consider that we have only two classes $c_1, c_2 \in C$.

We want to define what is the probability of $p(y|X)$.

To do that, we want to define what is the probability for just one instance:

$$p(y_i = c_1 | x_i)$$

As we have 2 classes, we can consider that this probability follow a Bernoulli law. So $y_i = 1$ if $y_i = c_1$ and $y_i = 0$ if $y_i \neq c_1$.

The probability to have 1 is given by a function that depends on x_i . So we have:

$$p(y_i = 1 | x_i) = g(x_i) \tag{1}$$

with $g(x_i)$ which give us a scalar from the vector of features x_i .

So it means that we have:

$$p(y_i | x_i) = \begin{cases} p(y_i = 1 | x_i) & \text{if } y_i = c_1 \\ 1 - p(y_i = 1 | x_i) & \text{if } y_i \neq c_1 \end{cases} \tag{2}$$

We can write the equation 2 like this:

$$p(y_i | x_i) = p(y_i = 1 | x_i)^{y_i} (1 - p(y_i = 1 | x_i))^{1-y_i} \tag{3}$$

But we want to know what is the function $g(x_i)$.

According to Bayes theorem, we have:

$$\begin{aligned}
p(y_i = 1|x_i) &= \frac{p(x_i|y_i = 1)p(y_i = 1)}{p(x_i)} \\
&= \frac{p(x_i|y_i = 1)p(y_i = 1)}{p(x_i|y_i = 1)p(y_i = 1) + p(x_i|y_i = 0)p(y_i = 0)} \\
&= \frac{\frac{p(x_i|y_i=1)p(y_i=1)}{p(x_i|y_i=0)p(y_i=0)}}{\frac{p(x_i|y_i=1)p(y_i=1)}{p(x_i|y_i=0)p(y_i=0)} + 1} \\
&= \frac{\frac{p(x_i|y_i=1)p(y_i=1)p(x_i)}{p(x_i|y_i=0)p(y_i=0)p(x_i)}}{\frac{p(x_i|y_i=1)p(y_i=1)p(x_i)}{p(x_i|y_i=0)p(y_i=0)p(x_i)} + 1} \\
&= \frac{\frac{p(y_i=1|x_i)}{p(y_i=0|x_i)}}{\frac{p(y_i=1|x_i)}{p(y_i=0|x_i)} + 1}
\end{aligned} \tag{4}$$

In probability, if q is the probability of an event, so the probability that this event is happening is given by $\frac{q}{1-q}$. This ratio is called the odds.

In our case, as $p(y_i = 1|x_i)$ is the probability that the event is happening, and $p(y_i = 0|x_i)$ is the probability that the event is not happening, we have the odds defined like this:

$$Odds = \frac{p(y_i = 1|x_i)}{1 - p(y_i = 1|x_i)} \tag{5}$$

In logistic regression, we want that the logarithm of the odds is linear. So according to equation 5, we want that:

$$\begin{aligned}
&\log \left(\frac{p(y_i = 1|x_i)}{p(y_i = 0|x_i)} \right) = w_0 + w_1 \cdot x_{i1} + \dots + w_d \cdot x_{id} \\
\Leftrightarrow \quad &\log \left(\frac{p(y_i = 1|x_i)}{p(y_i = 0|x_i)} \right) = w_0 + x_i w \\
\Leftrightarrow \quad &\log \left(\frac{p(y_i = 1|x_i)}{p(y_i = 0|x_i)} \right) = x_i w \quad \text{if } x = [1 \quad x_{i1} \quad \dots \quad x_{id}] \\
&\quad \text{and } w^T = [w_0 \quad \dots \quad w_d] \\
\Leftrightarrow \quad &\frac{p(y_i = 1|x_i)}{p(y_i = 0|x_i)} = e^{x_i w}
\end{aligned} \tag{6}$$

So according to the result of the equation 6, we can rewrite the equation 4 like this:

$$\begin{aligned}
p(y_i = 1|x_i) &= \frac{\frac{p(y_i=1|x_i)}{p(y_i=0|x_i)}}{\frac{p(y_i=1|x_i)}{p(y_i=0|x_i)} + 1} \\
&= \frac{e^{x_i w}}{e^{x_i w} + 1} \\
&= \frac{1}{1 + e^{-x_i w}} \\
&= \sigma(x_i w) \quad \text{with } \sigma(z) = \frac{1}{1 + e^{-z}}
\end{aligned} \tag{7}$$

σ in 7 is called the sigmoid function. (As e^{-x} is always negative or equal to zero, σ is always between 0 and 1)

So we can write the equation 3 like this:

$$\begin{aligned}
p(y_i|x_i) &= p(y_i = 1|x_i)^{y_i} (1 - p(y_i = 1|x_i))^{1-y_i} \\
&= \sigma(x_i w)^{y_i} (1 - \sigma(x_i w))^{1-y_i}
\end{aligned} \tag{8}$$

2.1.1 Cost function

Now, we want to define a cost function to optimize the parameter w .

Note that $p(y_i|x_i)$ is called the likelihood.

In fact, we want to maximise the probability of $p(y_i = 1|x_i)$ when $y_i = 1$ and maximise the probability of $(1 - p(y_i = 1|x_i))$ when $y_i \neq 1$.

So we want to maximize the likelihood, which is given by equation 8. And we want to maximize the probability for all instances and not only for instance i .

So we want to maximize:

$$\max \arg_w \sum_i^n p(y_i|x_i) \tag{9}$$

We want to use the gradient descent. So we want to have something to minimise. As we have to maximize the equation 8, we only have to invert this equation to have a minimisation problem.

We can easily see that the likelihood is always non zeros: we know that the exponential function is strictly positive, and according to equation 7, we have:

$$\begin{aligned}
0 &< e^{x_i w} < 1 + e^{x_i w} \\
\Leftrightarrow 0 &< \frac{e^{x_i w}}{1 + e^{x_i w}} < 1 \\
\Leftrightarrow 0 &< \sigma(x_i w) < 1 \\
\Leftrightarrow 0 &< p(y_i|x_i) < 1
\end{aligned}$$

because else, it means that all the elements are in one class, so we don't have two classes and we don't need to make some classification.

So we have according to equation 9:

$$\begin{aligned}
\max \arg_w \sum_i^n p(y_i|x_i) &= \min \arg_w \frac{1}{\sum_i^n p(y_i|x_i)} \\
&= \min \arg_w \sum_i^n \frac{1}{\sigma(x_i w)^{y_i} (1 - \sigma(x_i w))^{1-y_i}} \\
&= \min \arg_w \log \left(\sum_i^n \frac{1}{\sigma(x_i w)^{y_i} (1 - \sigma(x_i w))^{1-y_i}} \right) \\
&= \min \arg_w \sum_i^n (-\log(\sigma(x_i w)^{y_i} (1 - \sigma(x_i w))^{1-y_i})) \\
&= \min \arg_w \sum_i^n (-y_i \log(\sigma(x_i w)) - (1 - y_i) \log(1 - \sigma(x_i w)))
\end{aligned} \tag{10}$$

In the equation 10, we can apply log to the equation because the log function is always positive and increasing, and since we have said that $p(y_i|x_i)$ is always positive.

So we have inverted the likelihood, and then we have apply log function to result. As the log of the invert is equal to negative log (example: $\log(\frac{1}{x}) = -\log(x)$), we call the cost function the negative log likelihood function.

As y has a size of $n \times h$, X a size of $n \times d$ and w a size of $d \times h$, we can write the cost function with matrix product:

$$\begin{aligned}
&\min \arg_w \sum_i^n (-\log(\sigma(x_i w)) y_i - \log(1 - \sigma(x_i w)) (1 - y_i)) \\
&= \min \arg_w (-y^T \log(\sigma(Xw)) - (1 - y^T) \log(1 - \sigma(Xw))) \\
&= NLL(X, y)
\end{aligned} \tag{11}$$

2.1.2 Gradient

Now we have to compute the derivate of the cost function 11.

First, we want to find the derivate of the sigmoid function.

We have:

$$\begin{aligned}
\frac{d\sigma(z)}{dz} &= ((1 + e^{-z})^{-1}) \\
&= -1 \cdot (-e^{-z}) \cdot (1 + e^{-z})^{-2} \\
&= \frac{e^{-z}}{(1 + e^{-z})^2} \\
&= \frac{1}{1 + e^{-z}} \cdot \frac{e^{-z}}{1 + e^{-z}} \\
&= \sigma(z) \frac{e^{-z}}{1 + e^{-z}} \\
&= \sigma(z) \frac{1 + e^{-z} - 1}{1 + e^{-z}} \\
&= \sigma(z) \left(\frac{1 + e^{-z}}{1 + e^{-z}} - \frac{1}{1 + e^{-z}} \right) \\
&= \sigma(z) \left(1 - \frac{1}{1 + e^{-z}} \right) \\
&= \sigma(z) (1 - \sigma(z))
\end{aligned} \tag{12}$$

So according to the chain rule, we have:

$$\begin{aligned}
\frac{\partial}{\partial w_k} NLL(X, y) &= \frac{\partial}{\partial w_k} (-y^T \log(\sigma(Xw)) - (1 - y^T) \log(1 - \sigma(Xw))) \\
&= -y^T \frac{\partial}{\partial w_k} \log(\sigma(Xw)) - (1 - y^T) \frac{\partial}{\partial w_k} \log(1 - \sigma(Xw)) \\
&= -y^T \frac{\partial \log(\sigma(Xw))}{\partial \sigma} \frac{\partial \sigma}{\partial Xw} \frac{\partial Xw}{\partial w_k} \\
&\quad - (1 - y^T) \frac{\partial \log(1 - \sigma(Xw))}{\partial (1 - \sigma(Xw))} \frac{\partial (1 - \sigma(Xw))}{\partial \sigma} \frac{\partial \sigma(Xw)}{\partial Xw} \frac{\partial Xw}{\partial w_k} \\
&= -y^T \frac{1}{\sigma(Xw)} \sigma(Xw) (1 - \sigma(Xw)) X_{:,k} \\
&\quad - (1 - y^T) \frac{1}{1 - \sigma(Xw)} (-1) \sigma(Xw) (1 - \sigma(Xw)) X_{:,k} \\
&= ((1 - y^T) \sigma(Xw) - y^T (1 - \sigma(Xw))) X_{:,k} \\
&= (\sigma(Xw) - y^T \sigma(Xw) - y^T 1_{n,1} + y^T \sigma(Xw)) X_{:,k} \\
&= \left(\sigma(Xw) - \sum_i^n y_i \right) X_{:,k}
\end{aligned} \tag{13}$$

So the gradient of the cost function give us:

$$\nabla NLL(X, y) = \begin{bmatrix} \frac{\partial}{\partial w_0} NLL(X, y) \\ \vdots \\ \frac{\partial}{\partial w_d} NLL(X, y) \end{bmatrix}$$

Now, we have to make a gradient descent:

$$w^{i+1} = w^i - \eta \nabla NLL(X, y) \quad (14)$$

with η the learning rate.

2.2 Multinomial case

We have h classes $c_1, \dots, c_h \in C$.

If $y_i = c_k \in C$, we define $\tilde{y}_i$ a vector of size $1 \times h$ like this:

$$\tilde{y}_{ij} = \begin{cases} 1 & \text{if } j = k \\ 0 & \text{else} \end{cases} \quad (15)$$

So we have for instance $\tilde{y}_i = [0 \ 0 \ 1 \ 0]$ if $h = 4$.

Now, we note $y_i = \tilde{y}_i$. So y is of size $n \times h$.

We have for each $k \in [1, h]$ that the probability that $y_{ik} = 1$ or 0 depends on x_i .

So we have:

$$p(y_{ik}|x_i) = h_k(x_i) \quad (16)$$

Thus, we have:

$$p(y_i|x_i) = \prod_{k=1}^h p(y_{ik}|x_i)^{y_{ik}} \quad (17)$$

But what is the function $h_k(x_i)$? Is it the sigmoid function ?

No, it's not the sigmoid function, because this function work only for two variables. So we want a function like the sigmoid function that works for more than two variables, that's why, we'd try to generalize the sigmoid function to more than one variable

If we remember the equation 7, we have (binary case):

$$p(y_i = 1|x_i) = \frac{e^{x_i w}}{e^{x_i w} + 1} \quad (18)$$

Now, we suppose that we have

$$W = [w_1 \quad \cdots \quad w_h] \quad (19)$$

with each parameter w_k defined specially for the class k . (Note that the size of w_k is $d \times 1$, so the size of W is $d \times h$)

In binary case, we can define $W = [w_1 \quad w_2]$, with w_1 a column vector of 0.

So we can write the equation 18 like this:

$$\begin{aligned} p(y_i = 1|x_i) &= \frac{e^{x_i w_2}}{e^{x_i w_2} + e^0} \\ &= \frac{e^{x_i w_2}}{e^{x_i w_2} + e^{x_i w_1}} \\ &= \frac{e^{x_i w_2}}{\sum_{k=1}^2 e^{x_i w_k}} \\ &= \frac{e^{x_i w_2}}{\sum_{k=1}^h e^{x_i w_k}} \end{aligned} \quad (20)$$

So if we have h classes, we can define a function like this:

$$\begin{aligned} p(y_{ik} = 1|x_i) &= \frac{e^{x_i w_k}}{\sum_{j=1}^h e^{x_i w_j}} \\ &= \frac{e^{x_i w_k}}{\sum_{j=1}^h (e^{x_i W})_{1,j}} \\ &= \text{softmax}(W, x_i, k) \end{aligned} \quad (21)$$

We can easily see that softmax is between 0 and 1: As the exponential function is always positive, we have that:

$$\begin{aligned} 0 &< e^{x_i w_k} < e^{x_i w_k} + \sum_{j \neq k}^h e^{x_i w_j} \\ \Leftrightarrow 0 &< e^{x_i w_k} < \sum_{j=1}^h e^{x_i w_j} \\ \Leftrightarrow 0 &< e^{x_i w_k} < \sum_{j=1}^h e^{x_i w_j} \\ \Leftrightarrow 0 &< \frac{e^{x_i w_k}}{\sum_{j=1}^h e^{x_i w_j}} < 1 \\ \Leftrightarrow 0 &< \text{softmax}(W, x_i, k) < 1 \end{aligned} \quad (22)$$

So thanks to equation 21, we have:

$$\begin{aligned}
 p(y_i|x_i) &= \prod_{k=1}^h p(y_{ik} = 1|x_i)^{y_{ik}} \\
 &= \prod_{k=1}^h \text{softmax}(W, x_i, k)^{y_{ik}}
 \end{aligned} \tag{23}$$

2.2.1 Cost function

Now, as for the logistic regression, we want to find a cost function that maximise $p(y_i|x_i)$ for each instance. So we want to maximise the likelihood. According to equation 21 and equation 22, we have that $p(y_i|x_i) \neq 0$.

And we also want to use the gradient descent to optimize the parameter W , so we need a cost function to minimise.

We want to use the logarithm likelihood, because we have exponentials in softmax, so it will facilitate the derivate of the cost function. As we use the likelihood and as we want to maximise this function, we want to minimise the inverse of likelihood. However, we have $\log\left(\frac{1}{f(x)}\right) = -\log(f(x))$.

So we define the cost function as the negative logarithm likelihood function, and we want to minimise this function for all the instances.

So we have:

$$\begin{aligned}
\arg \min_W NLL(W, X, y) &= \arg \min_W \sum_i^n NLL(W, x_i, y_i) \\
&= \arg \min_W \sum_i^n -\log(p(y_i|x_i)) \\
&= \arg \min_W - \sum_i^n \log \left(\prod_{k=1}^h \text{softmax}(W, x_i, k)^{y_{ik}} \right) \\
&= \arg \min_W - \sum_i^n \sum_{k=1}^h \log (\text{softmax}(W, x_i, k)^{y_{ik}}) \\
&= \arg \min_W - \sum_i^n \sum_{k=1}^h y_{ik} \log (\text{softmax}(W, x_i, k)) \\
&= \arg \min_W - \sum_i^n \sum_{k=1}^h y_{ik} \log \left(\frac{e^{x_i w_k}}{\sum_{j=1}^h (e^{x_i W})_{1,j}} \right) \\
&= \arg \min_W - \sum_i^n \sum_{k=1}^h y_{ik} \left(x_i w_k - \log \left(\sum_{j=1}^h (e^{x_i W})_{1,j} \right) \right) \\
&= \arg \min_W \sum_i^n \sum_{k=1}^h y_{ik} \left(\log \left(\sum_{j=1}^h (e^{x_i W})_{1,j} \right) - x_i w_k \right) \\
&= \arg \min_W \sum_k^h y_{:,k}^T (\log (e^{XW} 1_{h,1}) - X w_k)
\end{aligned} \tag{24}$$

We want to explain how we transform the function $NLL(W, X, y)$ to a matrix product. We use $1_{x,y}$ to indicate that we use a matrix of ones of size $x \times y$. Note that we have W defined in equation 19 and that:

Matrix	Size
X	$n \times d$
y	$n \times h$
W	$d \times h$

So if we look at the sizes, we have:

$$\begin{aligned}
NLL(W, X, y) &= y_{:,k}^T (\log(e^{XW} 1_{h,1}) - Xw_k) \\
&= (n \times 1)^T \times (n \times d \times d \times h \times h \times 1 - n \times d \times d \times 1) \\
&= 1 \times n \times (n \times 1 - n \times 1) \\
&= 1 \times n \times n \times 1 \\
&= 1 \\
&= \text{scalar}
\end{aligned}$$

2.2.2 Gradient

We want to find the gradient of the NLL function obtained in equation 24.

We have for one instance:

$$\begin{aligned}
\frac{\partial}{\partial w_s} NLL(W, x_i, y_i) &= \frac{\partial}{\partial w_s} \sum_{k=1}^h y_{ik} \left(\log \left(\sum_{j=1}^h e^{x_i w_j} \right) - x_i w_k \right) \\
&= \sum_{k=1}^h y_{ik} \left(\frac{\partial \log \left(\sum_{j=1}^h e^{x_i w_j} \right)}{\partial \sum_{j=1}^h e^{x_i w_j}} \frac{\partial \sum_{j=1}^h e^{x_i w_j}}{\partial w_s} - \frac{\partial}{\partial w_s} x_i w_k \right) \\
&= \sum_{k=1}^h y_{ik} \left(\frac{x_i e^{x_i w_s}}{\sum_{j=1}^h e^{x_i w_j}} - y_{is} x_i \right) \\
&= \sum_{k=1}^h y_{ik} x_i \left(\frac{e^{x_i w_s}}{\sum_{j=1}^h e^{x_i w_j}} - y_{is} \right) \\
&= \sum_{k=1}^h y_{ik} x_i (\text{softmax}(W, x_i, s) - y_{is})
\end{aligned}$$

Here, we have the term $-x_i$ only if $s = t$ for $y_{it} = 1$, that's why I make $y_{is} x_i$ because $y_{is} = 1$ only if $s = t$ because of the equation 15 which define y .

So for all instances, we have:

$$\begin{aligned}
\frac{\partial}{\partial w_s} NLL(W, X, y) &= \sum_i^n \frac{\partial}{\partial w_s} NLL(W, x_i, y_i) \\
&= \sum_i^n \sum_{k=1}^h y_{ik} x_i (\text{softmax}(W, x_i, s) - y_{is}) \\
&= \sum_{k=1}^h y_{:,k}^T (X \odot (\text{softmax}(W, X, s) - y_{:,s}))
\end{aligned}$$

With $\odot$ a special product between a matrix A of size $n \times d$ and a vector B of size $n \times 1$, defined like this:

$$(A \odot B)_{ij} = A_{ij} \cdot B_i \quad (25)$$

So we define the gradient like this:

$$\nabla_W NLL(W, X, y) = \left[\frac{\partial}{\partial w_1} NLL(W, X, y) \quad \cdots \quad \frac{\partial}{\partial w_h} NLL(W, X, y) \right]$$

So we can make a gradient descent to find the optimal W :

$$W^{i+1} = W^i - \eta \nabla_W NLL(W, X, y) \quad (26)$$

with η the learning rate.